

JalSakshi

Clear water. Accountable tomorrow.

A worrying water test should end in a verified fix and a told resident, not in a notebook.

India's community water testing finds problems, but official audits show the follow-up routinely breaks down between the test, the lab, the repair and the people drinking the water. JalSakshi is an offline-first field app and case system that turns each flagged screening result into one owned case, which cannot close until there is a verified lab result, a recorded fix, a same-source retest and a resident update.

What exists today

Android app (v2.2 APK) + API on hosted staging. Works end to end on synthetic data.

What is not claimed

No real kit, lab partner, pilot or customer yet. Not a potability test.

What we show at this review

Offline capture that survives a crash; a closure the server refuses without evidence.

Team

Three builders: product, ML/native, backend/operations.

The follow-up after a bad test is where accountability is weakest, and that is the one step JalSakshi puts in charge

The problem

States reported **93.84 lakh** water samples tested with field kits in 2024-25^[1]. Official policy says an adverse kit result must go to a lab and trigger corrective action. In audited states that often did not happen. Karnataka retested **18%** of kit-contaminated samples in 2023-24 and **none** in the two years before^[2]. In Kerala, district labs "*did not communicate the test results*" to panchayats for remedial action^[3].

Who is affected

Residents drinking the water, most of all children and patients: only **72.8%** of tap samples at schools, anganwadis and health centres passed microbiological tests (2024)^[4]. Also the testers, whose work goes unanswered, and the operators who are answerable when an outbreak follows.

The failure in the current process

A flagged result has no single owner, deadline or closure rule. It moves on paper, phone calls and portals. Lab reports arrive after **9 to 106 days** against a 1–2 day norm^[2]. Repairs are not recorded against the result, and residents are often never told.

What JalSakshi changes

Each flagged screening becomes **exactly one case** with an owner and a due date. A lab result counts only after a **different person verifies it**. The server **refuses closure** until there is a verified report, a recorded corrective action, a later retest of the same source and a resident communication. Field capture works offline and survives crashes.

Most defensible differentiator

Not offline forms (mWater and ODK already do that well) and not phone colour-reading (prior art exists). The difference is **evidence rules enforced by default** across the whole chain, from screening to resident update. They are tested in code, and the app refuses false certainty. This is a combination claim from a dated desk review, not a claim to be first.

Build status, honestly

Verified (synthetic) offline capture, idempotent sync, guarded case workflow, resident complaint API, hosted staging, CI. **Partial** physical-phone runs, real kit, supervisor board UI, resident app screens. **Not production-cleared**: 10 named blockers^[25].

The next proof point

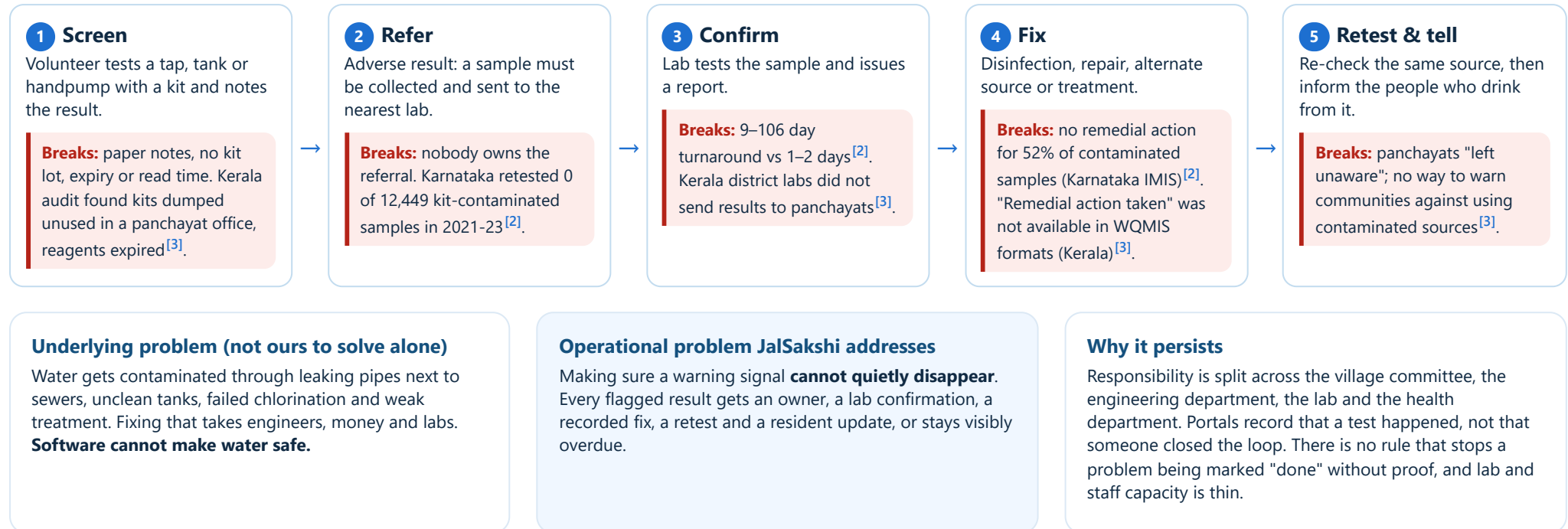
Live at this review: a record captured in airplane mode survives a forced restart and syncs once. Then the server refuses to close its case until the evidence exists. Next: two cold runs on a physical Android phone, a real kit protocol reviewed by a domain expert, and a 50–100 record shadow pilot with one campus operator.

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steps of the recorded end-to-end journey test pass on synthetic data (team record, 25 Sep 2026)

Testing finds contamination. The handoffs after the test are where it gets lost.

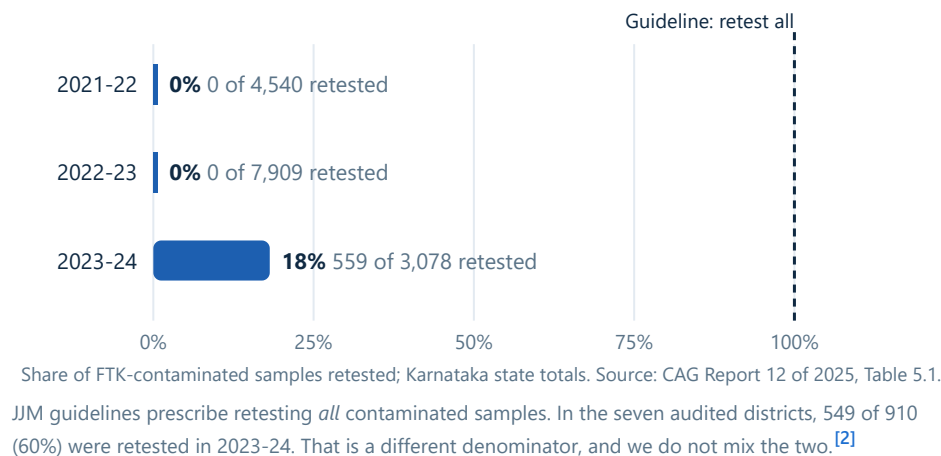
Under Jal Jeevan Mission, trained village volunteers (mostly women) test water with field test kits (FTKs). Those results are indicative screening only: an adverse result must go to a lab for confirmation and then to corrective measures^[1]. Each arrow below is a handoff between different people. The red notes are documented breaks.



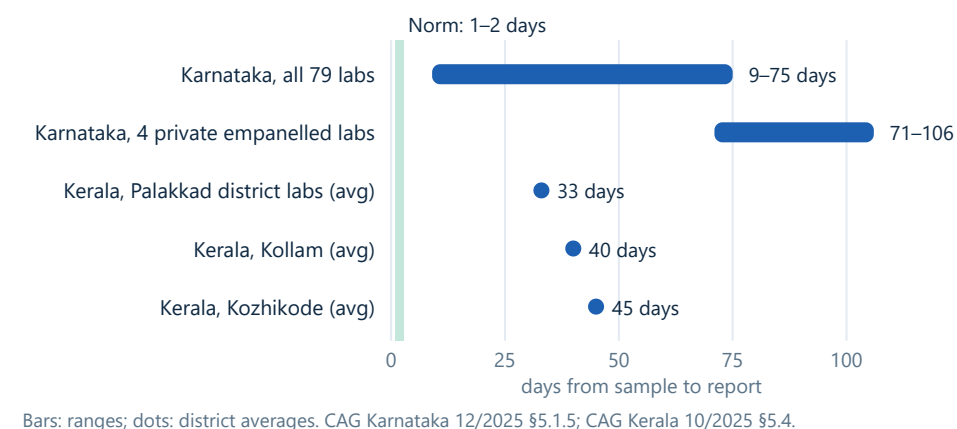
Two 2025 CAG audits document the same break: results that are not re-checked, not reported and not acted on

These are state findings (Karnataka and Kerala, JJM 2019-24), not national estimates. They are the most rigorous public evidence we found on what happens *after* a test, and both were tabled in early 2026.

Karnataka: almost no kit-flagged samples were re-checked



Lab confirmation takes weeks, not the 1–2 days the framework sets



"Audit however observed that the district laboratories did not communicate the test results to DP or GP for remedial action. This led to the GPs being left unaware of the status of water quality in their jurisdictional area."

CAG Kerala, Report 10 of 2025, para 5.1.2^[3]. The State replied that results were on WQMIS; audit noted WQMIS lacked sampling date, report date, location and remedial action taken.

2 of 28

test-checked Karnataka villages where the audit's own lab tests met the drinking-water standard^[2]

Small purposive sample, so not a prevalence estimate. In five villages the source was clean but household taps were not, which points to contamination during distribution.

What this means for JalSakshi

The gap is **not a lack of testing apps**. It is the absence of an owned, time-bound, evidence-checked path from a flag to a fix. That is a workflow and accountability problem, and software *can* make it visible, assignable and impossible to close falsely. It cannot add lab capacity.

When warnings do not become owned action, people fall ill, and some die

Four documented incidents across a city ward, an apartment complex, a village and a medical-college hostel. We separate confirmed findings from allegations. The last row of each card is **our analytical assessment**, not a claim that JalSakshi would have prevented the harm.

Bhagirathpura, Indore

DEC 2025 – FEB 2026 · MUNICIPAL SUPPLY

What	Sewage entered a Narmada drinking-water line; a leak was found where a toilet had been built over the pipe ^[8] .
Confirmed	36 deaths reported 26 Dec – 25 Feb. A High Court judicial commission (set up 27 Jan 2026 ^[7]) reportedly linked 24 to contaminated supply and held six officials responsible. The report is confidential ^[6] .
Alleged	Residents' complaints to the municipal helpline went unattended for days from 25 Dec ^[8] .
Failure	Delayed pipeline replacement; complaints not converted into an owned investigation ^[6] .
Relevance	Complaint intake with a reference number, and illness reports that alert a named supervisor, would make the warning visible and dated. The pipe still needed replacing.

DLF flats, Kakkanad, Kochi

MAY – JUN 2024 · PRIVATE APARTMENT COMPLEX

What	Diarrhoea outbreak after mixed-source water (KWA, borewells, rainwater, tankers) was pumped to towers ^[11] .
Confirmed	Health department identified 441 ill in two weeks, 5 hospitalised; E. coli confirmed in samples from the complex's tank ^[11] .
Alleged	A lab report dated 29 May showed E. coli but was not shared with residents or officials until 13 June ^[10] .
Failure	A positive result did not trigger action or disclosure.
Relevance	An adverse lab result attached to an open case, with closure blocked until residents are informed, targets exactly this gap. It depends on the operator using the system.

Kavadigarahatti, Chitradurga

JUL – AUG 2023 · VILLAGE OVERHEAD TANK

What	Residents fell ill after drinking from the only overhead tank, which served about 220 houses ^[13] .
Confirmed	4 of 5 samples unfit; <i>Vibrio cholerae</i> found in tank and tap water ^{[12][13]} . Five to six deaths reported; officials suspended.
Disputed	Cause: neglected tank cleaning vs. alleged deliberate contamination; officials rejected the latter ^[13] .
Failure	Routine tank sanitation and source checking were not evidenced.
Relevance	Source history and scheduled same-source retests make a missed check visible. They cannot prevent deliberate tampering.

Gandhi Medical College hostels, Hyderabad

APR 2026 · CAMPUS

What	Students reported worms in the hostel drinking-water tank ^[14] .
Confirmed	30+ of ~300 undergraduate students with diarrhoea treated; one hepatitis case; samples sent to the Institute of Preventive Medicine ^[14] .
Alleged	Students say they raised the issue earlier and cleaning began only once illness was severe ^[14] .
Failure	A known concern with no dated owner or deadline.
Relevance	This is our first target segment: campus operators answerable for their own tanks. A dated complaint and an overdue case are hard to ignore.

We name these places to show what is at stake, not to assign blame beyond the cited findings. Death counts differ between official and resident accounts; we report the figures attributed to the cited sources.

The exposure is national, the testing workforce already exists, and people cannot tell unsafe water by looking

All figures below come from the Ministry of Jal Shakti's own data. The functionality assessment is an independent national survey: 19,401 villages and 2,32,691 households, fieldwork July–October 2024 (monsoon), villages already declared "Har Ghar Jal" only^[4].

76.0%

of household tap samples passed microbiological tests. About **1 in 4** did not^[4]

72.8%

passed at **schools, anganwadis and health centres**^[4]

92.4%

of households said they were **satisfied** with water quality. Perception is not a safety test^[4]

27.2%

of villages had a field test kit available on survey day; 55.2% had a water committee^[4]

93.84 lakh

FTK tests in 2024-25 (47.59 lakh in 2025-26 to 12 March; a partial year, not a decline)^[1]

24.80 lakh

women trained to test with FTKs (cumulative, as of 12 Mar 2026)^[1]

2,870

water-quality labs reported by states (4 Feb 2026). Results are shown to citizens on the JJM dashboard^[5]

9–106 days

lab turnaround observed in Karnataka vs a 24–48 hour norm^[2]

What these numbers mean for us

Contaminated water reaches people even where taps exist, and residents cannot see it. So **each flagged test matters**, and a flag that nobody follows up is a missed warning. We do not extrapolate state audit percentages to India.

Why now

The workforce, kits and labs exist, and tests run at crore scale each year. Audits tabled in **Feb–Mar 2026** name the weak step precisely. Low-cost Android phones can now store evidence offline and run small models locally: our 896-byte model runs in 1.7 ms on an emulator.

Why it is practical

JalSakshi does not need new infrastructure. It wraps the kit test and lab report that already happen, and the most expensive step (lab confirmation) is already funded by the system. What it adds is ownership, deadlines and proof.

Good tools exist for recording tests. Few enforce, by default, that a flagged result ends in verified action.

Scope: desk review of official product pages and documentation, 21–25 Sep 2026. No hands-on trial of these tools yet. "?" means **not verified**, not "absent". • built in • possible by configuration or partial • not provided.

ALTERNATIVE	TYPE	WORKS OFFLINE IN THE FIELD	KIT PROTOCOL: LOT, EXPIRY, READ WINDOW	FLAG OPENS AN OWNED CASE WITH DUE DATE	LAB RESULT VERIFIED BY A SECOND PERSON	CLOSURE BLOCKED WITHOUT LAB + FIX + RETEST + RESIDENT UPDATE	RESIDENT COMPLAINT → SAME CASE	COST TO OPERATOR
Paper / WhatsApp + lab	Current manual process	•	•	○	•	○	○	Low cash, high chasing time
JJM-WQMIS portal + app ^{[18][5]}	Government system; integration partner	?	?	?	?	?	• results shown publicly	Free (government)
mWater Surveyor + Workflows ^[15]	Direct / adjacent competitor	•	•	• assign, steps, overdue alerts	•	•	?	Platform free; paid services
ODK Collect/Central + Entities ^[16]	Adjacent; configurable foundation	• (2024.3+)	•	• case management	•	•	?	Self-host, or Cloud from USD 199/mo
Akvo Caddisfly ^[17]	Adjacent: phone kit reading	?	•	?	?	?	?	Open source (GPL-3.0)
JalSakshi	Our product	• verified	• synthetic kit	• verified	• verified	• verified	• API only	Proposed ₹7,500/mo per campus

Where others are strong

mWater and ODK are mature, offline-first and free or cheap, and either can be *configured* into much of this loop. Phone-based colour reading is prior art (Akvo Caddisfly; peer-reviewed work including fully on-device systems^[22]^[23]). WQMIS is the official record and the right integration target.

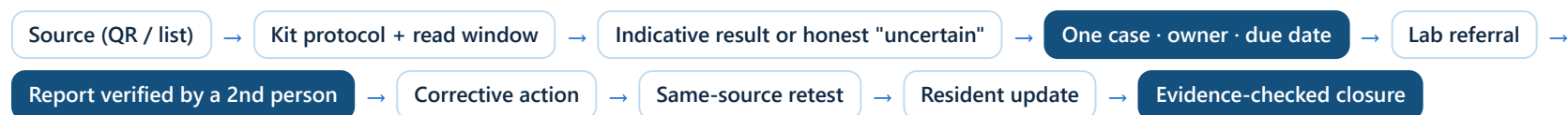
The underserved need

Operators who are **answerable for a set of sources** (campuses, hostels, programmes) but lack a data team to design workflows need a loop that is **correct out of the box**: evidence rules they cannot skip, screening kept separate from lab confirmation, and residents inside the same record.

How we will test this claim

Before building further we will run a time-boxed mWater/ODK sandbox trial against the same pilot workflow (team gate T02). If configuration covers it, JalSakshi becomes a capture-and-evidence companion that integrates, rather than a replacement.

JalSakshi turns a flagged result into one owned case that can only close on evidence



PROBLEM (FROM P.3)	CAPABILITY	USER ACTION	EXPECTED OPERATIONAL BENEFIT	VALIDATION METHOD
1 · Results on paper, no kit lot or timing	Protocol-bound capture; lot/expiry check; two-clock read timer; offline receipt	Worker scans source, follows timer, records reading or marks uncertain	Complete, dated, attributable records even without signal	Record completeness in pilot vs paper baseline
2 · Referral has no owner	Flagged sample opens exactly one case; owner and due date	Supervisor assigns and refers to lab	No unowned flags; overdue visible	% flags owned within 24 h; median flag → referral
3 · Lab result slow, unshared or unverified	Lab report recorded; verification by a different person; supersession	Supervisor records; lab reviewer verifies	Only checked results drive decisions; delay is measured	Referral → verified-report days, per lab
4 · Fix not recorded	Corrective-action record with completion evidence	Operator logs chlorination, repair or alternate source	"Done" means something specific	% cases with action evidence
5 · No retest; residents not told	Same-source later retest; communication record; closure guard; complaint intake	Worker retests; supervisor records the notice; residents report and track by reference	No silent closures; residents have a route in	0 closures without evidence; complaint → case linkage

Who uses it

Field worker / volunteer: Android app, offline.
Supervisor: triage, referral, verification, closure. **Lab reviewer:** verifies reports. **Operator:** records fixes.
Resident: complaints, status, notices.

What triggers it

A scheduled source test, a kit reading needing review, or a resident's complaint (including an illness report).

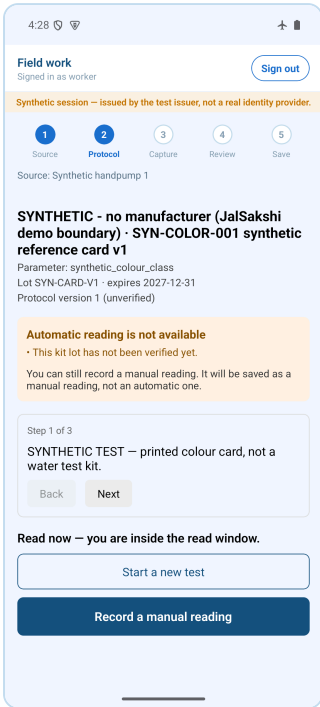
Where our responsibility ends

JalSakshi does not treat water or certify it potable, and it does not replace lab testing. It will never show "safe to drink". It cannot force staff to act. It makes inaction **visible, dated and attributable**, and the operator stays accountable.

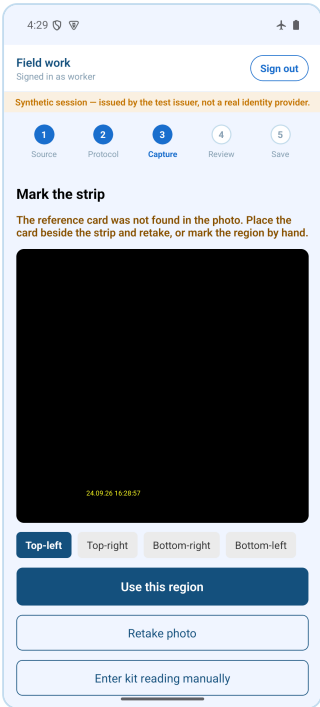
HOW SIMPLY IT WORKS

Seven steps from a doubtful reading to a closed case, with real screens from the working build

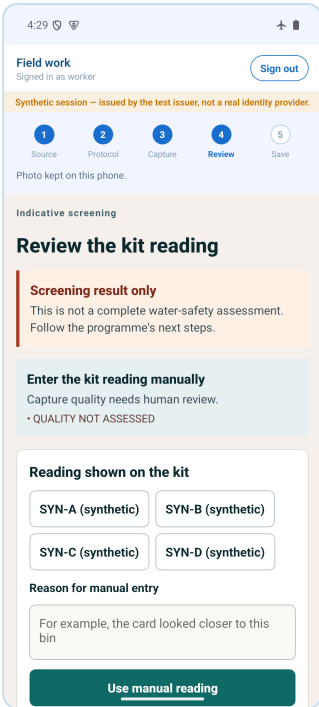
Illustrative scenario, not a real event: a hostel caretaker tests a rooftop tank on the monthly round, with no signal on the roof. All screens below are genuine Android-emulator captures of JalSakshi v2 using the **synthetic** test card SYN-COLOR-001. They are not mock-ups, and they are not real water results.



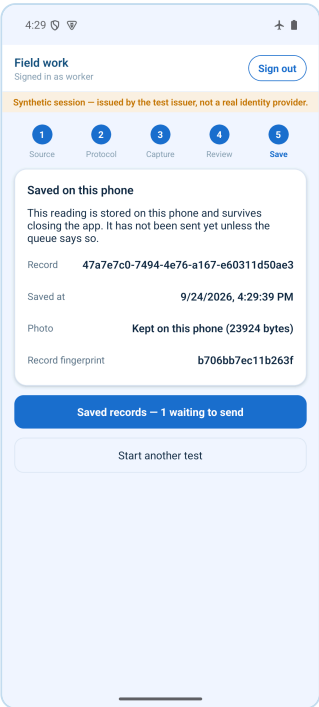
1 • Source & protocol
Kit lot and expiry are checked. The two-clock timer allows a reading only inside the read window. An unverified lot means manual-only.



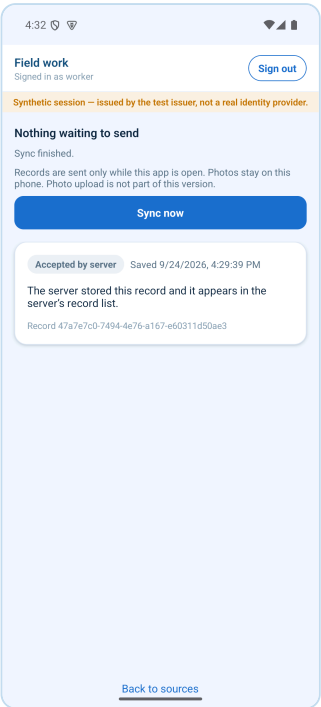
2 • Capture
Photo of strip and reference card. When the card is not found, the worker retakes or marks the region by hand.



3 • Review
"Screening result only." The quality gate failed, so no automatic reading: the worker enters it with a reason.



4 • Saved offline
Receipt with record ID and fingerprint. It survives a forced restart in airplane mode.



5 • Synced once
On reconnect the server accepts it. A re-sent copy is recorded as a duplicate, never a second case.

SERVER SIDE - NO UI IN THIS APP

6 • Case & lab. One case opens with an owner. The lab report counts only after a *different* reviewer verifies it (the recorder gets 403).

7 • Close. The first close attempt is **refused** and the checklist names the missing resident communication. After a same-source retest and a recorded notice, it closes.

Evidence: tests/e2e/field-case.spec.ts, 8/8 real HTTP steps, team record 25 Sep 2026. The supervisor board is a separate web app.

Decisions and handoffs

The worker decides the reading. The supervisor owns triage, referral and closure. A separate lab reviewer verifies, and the operator records the fix. Each step carries a name and timestamp.

Completion or escalation

A case closes only on complete evidence. Otherwise it stays open with its due date, and overdue cases are listed for the supervisor and the programme lead.

What the resident sees

A complaint reference number and status (API built). The public view shows only lab-verified values with lab name and date, never a screening result or "safe to drink".

The system is designed to refuse, queue or fall back to a human, never to guess

SITUATION	WHAT JALSAKSHI DOES	STATUS
No network in the field	Saves locally with a receipt; queues; syncs on reopen. Multi-person decisions wait for the server.	Verified (synthetic)
Phone killed mid-save	Record is either whole or absent, never half-written.	Verified (synthetic)
Response lost, request re-sent	Same content → "duplicate"; changed content → conflict. Never a second case.	Verified (synthetic)
Bad photo, missing reference card	Quality gate refuses an automatic reading; retake or manual entry with a reason.	Verified (synthetic)
Expired lot, reading outside window, clock tampered	Assisted reading blocked; manual path stays open; timing marked indeterminate.	Partial: synthetic kit
Model unsure or wrong	Model is research-only. It abstains, never shows confidence as "safety", and can be disabled remotely (kill switch).	Partial: research
Offline lease expired or phone clock rolled back	Access locks; saved records are kept (5 h rollback test: 0 lost).	Verified (synthetic)
Someone approves their own lab report	Refused at the API and the database.	Verified (synthetic)
Premature closure; two supervisors edit at once	Refused with the missing items named; stale edit gets a version conflict.	Verified (synthetic)
Server asleep (free hosting cold start ~1 min)	App survives cold start; field records wait in the queue. Always-on hosting is required before a pilot.	Partial
Resident has no smartphone; SMS alert needed	Alerts are queued, not delivered : no SMS, push or IVR provider yet. Printed or notice-board communication is recorded instead.	Roadmap

Feature inventory at a glance

Verified (synthetic)	17
Partial	12
Hackathon	1
Roadmap	6
Proposed	3

Counted from the full matrix in Appendix B (area → feature → user → problem → status → evidence). "Verified (synthetic)" means code exists and passes automated tests and/or emulator runs on synthetic data. Most still owe an independent review.

Not yet safe for real data

Ten production blockers are open, including database row-level security (the API uses the table owner), no phone encryption at rest, secrets to rotate, no licence, and an unsigned closure policy^[25]. The team's own release review lists them.

Built around field constraints: no signal, cheap phones, thin lab capacity and skeptical auditors

CRITERION	DESIGN CHOICE	TRADE-OFF WE ACCEPT
Fit with existing work	Wraps the kit test and lab report that already happen. Nothing replaces the lab. In a pilot the old process remains the record of decision (shadow mode).	Double entry during the pilot; value must show within weeks
Operating constraints	Android first, offline after provisioning; tiny on-device model (68 parameters, 896 bytes) with no per-image cloud fee; foreground sync instead of unreliable background jobs.	iOS later; multi-person decisions need connectivity
Reliability	Crash-safe local store, idempotent sync with content hashes, ordered change feed, rehearsed backup/restore and rollback.	Engineering effort spent on durability, not on dashboards
Accountability	Evidence rules live in the server's state machine and database constraints, not in the UI. Every decision has a name, a time and a version.	Stricter than some operators want: the closure policy needs domain sign-off
Honesty	Screening is always labelled "indicative". Confidence numbers are never shown. Public status never says "safe". Synthetic data is watermarked.	Less "impressive" output than a model claiming accuracy
Interoperability	Versioned OpenAPI contract with a generated client; safe CSV export. No claimed WQMIS integration until permission and an API exist.	Manual export to official portals for now
Cost & scale	One modular FastAPI service + PostgreSQL; pilot infrastructure estimated at USD 59–100/month ^[20] . Measured locally: 10 req/s at p95 50 ms, no lost or doubled records at 50 req/s ^[25] .	PostgreSQL throughput not yet load-tested
Maintenance	No LLM, vector store or microservices. Deterministic rules come first, and the model only where it beats them. CI on every push.	Fewer buzzwords

When a simpler tool is enough

A site with three taps and one diligent manager can use a register and a lab. JalSakshi earns its place when there are **many sources, several people and residents who need proof.**

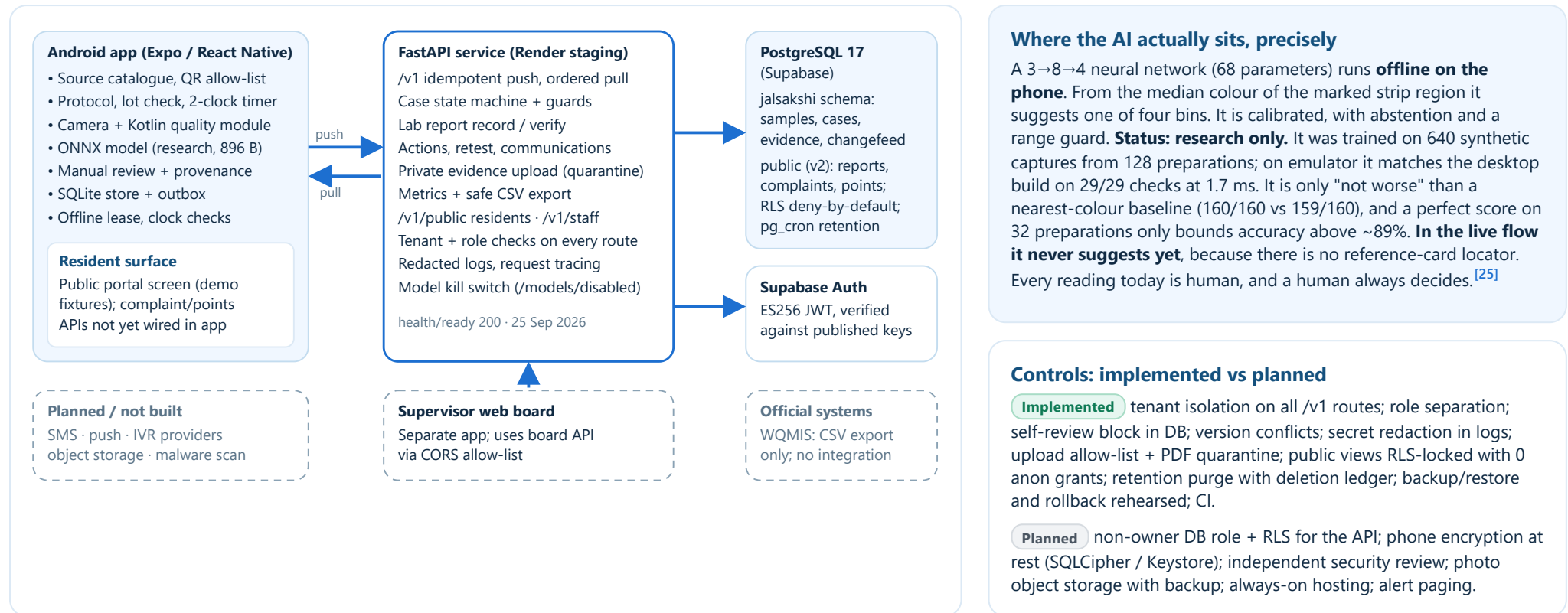
Advantage we have now

A tested evidence engine with offline durability, which is the part most teams skip. The code itself can be copied, so it is not a moat on its own.

Advantage that could develop

Validated kit-and-device data, a library of reviewed protocols, operator relationships and a track record that auditors trust. None exists yet. They are earned through pilots.

One offline-first app, one guarded API, one database: small enough to run, strict where it matters



Known gap: the mobile sync cases (jalsakshi schema) and the resident-layer reports (public schema) are two case models that are not yet bridged. We recommend unifying them before any pilot. Data flow: phone → push (idempotent) → case engine → change feed → phones and board. Evidence photos stay on the phone unless separately authorised.

The hard, invisible parts are built. What remains is real-world proof, and it needs partners, not just code.

PHASE	DELIVERABLE	DEPENDENCY	SUCCESS CRITERION	OWNER
Complete	Offline field slice (M1), evidence-to-action case loop API (M2), resident layer API (P1–P13, four items still partial), staging, CI, backup/rollback rehearsal	—	Emulator airplane-mode journey; 8/8 journey test; staging smoke 7/7	Team (roles A/B/C)
This hackathon	5-minute live story: offline capture → forced restart → sync once → refused closure → evidenced closure	Emulator or phone; staging awake	Every step shown live, or a labelled recording as backup	A (product)
Before next review (our suggestion)	Two cold runs on a physical Android phone; a minimal resident-complaint screen wired to the existing API; case timeline on the board	A phone + USB; board app access	Recorded runs with build hash; complaint appears as a case	A / C
Before a pilot	Real kit + protocol with a domain reviewer; signed closure policy; RLS + non-owner role; phone encryption; secret rotation; Hindi; licence; always-on hosting; one case model	Lab or domain partner; owner decisions	All 10 blockers in the release review closed with evidence	A + C; domain partner
Pilot	Shadow pilot: 50–100 records, one operator, 8–12 weeks	Operator agreement, consent, data ownership	Thresholds on p.14 met; buyer conversation recorded	A
Production	Release checklist: backup running, restore rehearsed on production, alerts watched 24 h	Pilot outcome	Owner approval recorded; production smoke passes	C
Expansion	Second kit/parameter, validated model on real preparations, SMS/IVR, official export	Real data, providers, permissions	Per-kit evaluation report; leave-phone-out results	B

Risk 1 · No domain or lab partner

Without one we cannot validate a kit or sign a closure policy. **Reduce:** approach a district PHED lab or university chemistry/microbiology department now. Pick one parameter first. Keep the model research-only until real preparations exist.

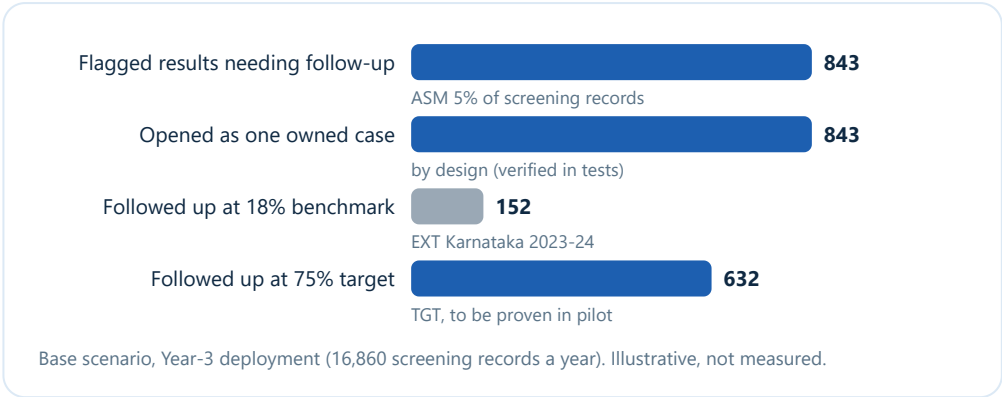
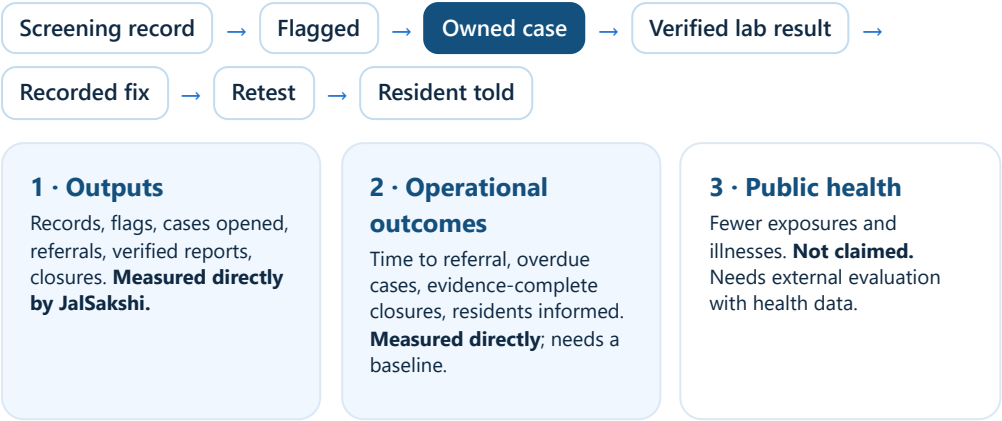
Risk 2 · Staff do not act on cases

Software makes delay visible; it does not create capacity. **Reduce:** pilot only with an operator who names an owner for follow-up. Measure time to referral and overdue cases, and design escalation with them.

Risk 3 · A free, configurable tool is "good enough"

mWater and ODK may cover the workflow. **Reduce:** run the configure-versus-build trial on the same pilot workflow before more feature work. If configuration wins, pivot to a companion or integration.

We claim what the product can count: owned cases and completed follow-ups. Health effects need independent evaluation.



PER YEAR, AT YEAR-3 DEPLOYMENT	LABEL	CONSERVATIVE	BASE	OPTIMISTIC
Deployment (Year-3 customers, Ess / Plus / Prog)	ASM	10 / 3 / 1	25 / 8 / 3	50 / 15 / 8
Water sources monitored	ASM	730	2,005	4,550
Screening records per year	ASM	6,360	16,860	35,400
Flagged results (5%)	ASM	318	843	1,770
Flagged results opened as one owned case	TGT	318	843	1,770
Followed up at 18% external benchmark	EXT	57	152	319
Follow-through target (lab-verified + retest ≤30 days)	TGT	50%	75%	90%
Additional documented follow-ups vs benchmark	TGT	102	481	1,274

Formula: records = Σ customers × sources × tests/year; flagged = records × 5%; follow-ups = flagged × follow-through rate. The 18% benchmark is Karnataka's 2023-24 retest rate^[2], a different context. **Our real baseline must be measured** in the pilot's shadow period.

Pilot measurement plan	
Baseline	4 weeks of shadow logging of the current process at the same sources: flags, referral dates, report dates, actions, notices
Period	8–12 weeks, 50–100 records, one operator, sources they already test
Metrics	% flags owned ≤24 h · median flag → referral (days) · referral → verified report (days, per lab) · % flagged with verified report + retest ≤30 days · closures without evidence · records needing correction
Thresholds TGT	≥95% owned ≤24 h · referral ≤2 days median · ≥75% follow-through · 0 evidence-less closures · <5% corrections
Reported with	Missing data, season, worker experience, kit lot; no causal claim from an uncontrolled before/after

SaaS fits because the value recurs every month: sources are tested, flags arise and evidence accumulates

SEGMENT	NEED	WILLING TO PAY	ABILITY TO ADOPT	SALES CYCLE	SETUP BURDEN	RANK
Residential campuses (universities, hostels, residential schools)	High: own tanks, answerable to students and parents	Medium: facilities budget	High: one decision-maker	1–3 months	Low	1
NGO / implementation partners running village surveillance	High: donor reporting	Medium: programme or CSR budgets	Medium: field training	3–6 months	Medium	2
Private testing labs	Medium: client portal	Medium	High	1–3 months	Low	3 (channel)
Housing societies / RWAs	Medium–high after incidents	Low: fragmented	Medium	Variable	Low	4
State PHED / district missions	High	High but slow	Low: WQMIS exists	12+ months, tender	High	Later: integrate, don't compete

Rankings are our judgement from the evidence, not buyer interviews. No buyer, customer or willingness-to-pay figure exists yet.

Who is who (campus segment)

End users: caretakers, estates staff, supervisors. **Beneficiaries:** students, staff, residents.
Paying customer: the institution. **Budget owner:** estates / facilities or the hostel warden's office. **Procurement:** registrar or purchase committee, often under a threshold for annual software. Residents never pay, and we never sell data.

Why a campus would pay rather than keep a register

It gets **proof** for parents, regulators and accreditation visits that every flagged tank was followed up. It gets fewer hours spent chasing labs and contractors, a resident channel that surfaces problems early, and a dated record if something goes wrong. The incidents on p.5 show the reputational and human cost of a known problem left unowned.

What could stop adoption

Free alternatives; extra work for caretakers; fear of creating a written record of problems; no lab contract in place; data-protection review.

How we validate willingness to pay

Ask 10 campus facilities heads for their current annual water-testing spend and staff time. Offer the pilot free, then quote a paid annual plan and record yes/no/why. Success = 2 signed letters of intent at the quoted price.

Charging unit: per site (and source band), not per user

Value and cost scale with the number of **water sources monitored**, not with users. Per-user pricing would penalise adding residents and volunteers, which is exactly who we want in the system.

Bottom-up reachable market (campus segment only; illustrative)

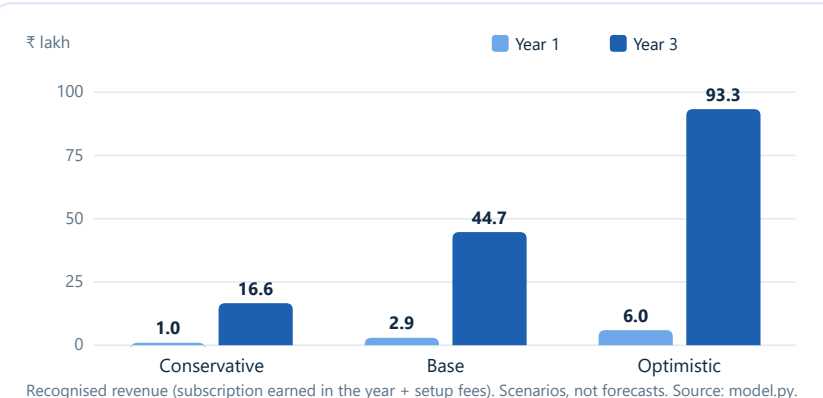
45,473 colleges registered (AISHE 2021-22)^[19] × 30% residential with their own storage (**ASM**) = 13,642 × 10% reachable in 2–3 states in three years (**ASM**) = **1,364 campuses**. At the Essentials price that is a ₹12.3 crore a year ceiling, not a forecast. The base Year-3 campus count is 2.4% of it. NGO programmes, labs and schools are not added, because their counts are unverified.

Pricing is a hypothesis. At these prices, break-even needs about 58 customers: the number to validate.

TIER	TARGET CUSTOMER	INCLUDED VALUE	BILLING UNIT	INDICATIVE PRICE (EXCL. GST)	SUPPORT BURDEN	EXPANSION PATH
Validation pilot	Operator new to JalSakshi	One site, ≤25 sources, 8–12 weeks shadow use, training, baseline report	Per pilot	₹0 (first two); later ₹25,000 credited on conversion	High (founder-led)	Converts to Essentials
Campus Essentials	Single campus / hostel group, residential school	Case workflow, offline field app, lab-report verification, resident complaint intake, exports	Per site / year	₹7,500/mo (₹0.9 L/yr) + ₹30,000 setup	Email, next business day	More sources → Plus
Campus Plus	Large or multi-block campus, hospital	Up to ~100 sources, lab-partner accounts, multiple supervisors, priority support	Per site / year	₹18,000/mo + ₹60,000 setup	Priority, 4 business hours	Group / multi-campus
Programme	NGO / implementation partner running village surveillance	Cluster up to ~300 sources, field-team offline leases, programme dashboards & exports	Per programme cluster / year	₹30,000/mo + ₹75,000 setup	Named contact, field refresher	More clusters / districts

Anchors: ODK Cloud Standard USD 199/mo ≈ ₹19,104^{[16][21]} (Essentials = 39% of it); mWater is free with paid services^[15]. **Outside subscription:** setup (one-time), training ₹15,000/day, integration (quoted), SMS/tags at cost (not modelled). Grants and prizes are **never counted as revenue**.

	CONSERVATIVE	BASE	OPTIMISTIC
Year 1 (Oct 2026 – Sep 2027; paying starts from month 7)			
Paying customers at year end (Ess / Plus / Prog)	2 / 0 / 0	3 / 1 / 0	5 / 1 / 1
Recognised revenue (subscr. + setup)	₹1.0 L (0.4 + 0.6)	₹2.9 L (1.4 + 1.5)	₹6.0 L (3.1 + 2.9)
Direct delivery cost (infra, hosting, onboarding)	₹1.8 L	₹2.3 L	₹3.4 L
Gross profit (margin)	–₹0.8 L (–82%)	₹0.6 L (22%)	₹2.6 L (44%)
Operating expenses (founders unpaid)	₹7.2 L	₹7.2 L	₹7.2 L
Operating result, founders unpaid	–₹8.0 L	–₹6.6 L	–₹4.6 L
ARR at end of Year 1 (MRR × 12)	₹1.8 L	₹4.9 L	₹10.3 L
Year 3 (Oct 2028 – Sep 2029; salaried team)			
Customers at start → end (all tiers)	6 → 14	16 → 36	29 → 73
Gross new customers (incl. churn)	9	22	47
Recognised revenue (subscr. + setup)	₹16.6 L (12.9 + 3.8)	₹44.7 L (35.7 + 9.0)	₹93.3 L (74.0 + 19.4)
Direct delivery cost (incl. customer success)	₹10.4 L	₹21.2 L	₹37.6 L
Gross profit (margin)	₹6.2 L (38%)	₹23.5 L (53%)	₹55.7 L (60%)
Operating expenses	₹41.0 L	₹59.0 L	₹81.0 L
Operating result	–₹34.8 L	–₹35.5 L	–₹25.3 L
ARR at end of Year 3	₹19.1 L	₹50.6 L	₹106.2 L
Break-even customers (Year-3 mix & costs)	≈ 42	≈ 58	≈ 77



Reading the numbers

Year-1 revenue (₹2.9 L base) is far below year-end ARR (₹4.9 L), because customers start mid-year. Year 1 looks small only because **founders are unpaid**; paying each of three founders ₹5 L would make the base result –₹21.6 L. In Year 3 the base case earns ₹44.7 L at 53% gross margin, and its operating result is still –₹35.5 L. It needs about 58 customers at the Year-3 mix and costs (it has 36). Each extra base-mix customer adds about ₹1.1 L a year of contribution.

Start with one campus that owns its tanks, prove the follow-through, then sell the evidence

First pilot

Ideal customer: a residential university or hostel group with 15–40 tanks, taps and filters, and an estates office that already pays for lab tests. **Scope:** one parameter, one kit, 8–12 weeks in shadow mode. **Decision-maker:** estates head or registrar. **Onboarding:** source QR tagging, a half-day training, a named follow-up owner. **Conversion:** the baseline-versus-pilot report, then an annual Essentials quote. **Expansion:** more blocks, then Plus; sister campuses.

Channels and partners to pursue (none confirmed)

University estates and facilities networks; district or private NABL labs (as referral partners and channel); NGOs working with Pani Samitis under JJM; campus incubators. Hackathon incubation support would be a grant, not revenue.

Procurement friction

Annual software approvals, data-protection review, and reluctance to create written records of problems. Mitigations: a low annual price under common purchase thresholds, a data-ownership clause and an export-anytime guarantee.

Demonstration at this review

Show now: sign in against the hosted API; airplane-mode capture; kill and reopen; sync once (duplicate refused); bad capture → manual reading; server refuses closure, naming the missing evidence; evidenced closure. **Judges should notice:** the refusal, the single record after a retry, and the "screening only" wording. **If a dependency fails:** offline steps need no network; server steps fall back to a labelled recording of the 8-step journey. Never a fabricated result.

Not interface-only: real persistence, real server guards, real sign-in and CI tests, all verifiable in the repository.

OFFICIAL CRITERION ^[24]	WHERE THIS DOCUMENT ANSWERS IT	STRONGEST EVIDENCE	HONEST LIMIT
Innovation	p.7, p.8, p.11	Evidence-enforced closure + protocol-bound, uncertainty-aware capture + resident channel, in one tested system	Combination claim; ingredients have prior art
Technical execution	p.9, p.10, p.12	Crash-safe offline store, idempotent sync, guarded state machine, on-device model, hosted staging, CI	Emulator only; independent review owed
Feasibility	p.11, p.13, p.16	Small stack, estimated USD 59–100/month pilot infrastructure; named blockers with owners	No kit, lab or operator partner yet
Impact	p.3–6, p.14	Government audits and survey data; measurable pilot metrics	No field data; no health claim
Presentation (theme: Project to Product)	p.15–17	Buyer, pricing unit, break-even and pilot path	Pricing and willingness to pay unvalidated

Every year, crores of water tests generate warnings. The ones that matter most are the ones nobody owns.

The cost of leaving the gap open

In audited Karnataka, most kit-flagged samples were never re-checked. In Kerala, panchayats were never told. In Indore, Kochi, Chitradurga and a Hyderabad hostel, warnings came before the illness. Each unowned flag is a warning that someone saw and nobody acted on.

Why it deserves validation

The weakest link is documented by the state's own auditors. The workforce, kits and labs already exist. The hard engineering (durability, sync, evidence rules) is built and tested. What is missing is real-world proof, and a pilot can supply it cheaply.

The change JalSakshi aims for

A flagged result that **cannot disappear**. It has an owner and a date. It closes only on a verified lab result, a recorded fix, a retest and a told resident, or it stays visibly overdue. Honest about uncertainty, and working without a signal.

What we will demonstrate next

- Two cold end-to-end runs on a physical Android phone, recorded with build hashes
- A real kit protocol reviewed by a lab or domain expert
- A shadow pilot with one campus operator: baseline vs JalSakshi on time-to-referral and evidence-complete closure
- A priced conversation with a real buyer, recorded as quotes, not projections

JalSakshi: the witness (*sakshi*) that makes sure a warning about water is followed through.

Ten questions we expect, with honest answers

1 · Why does this need to exist?

Two state audits show kit-flagged contamination is rarely re-checked, reported or fixed on record^{[2][3]}. Documented outbreaks follow the same pattern: an unanswered warning. Tools record tests; few enforce the loop.

2 · Why are existing systems insufficient?

WQMIS is the official record, but the Kerala audit found remedial action was not captured in its formats. mWater and ODK can be configured, but that needs expertise, and they do not enforce evidence rules by default. We have not trialled them hands-on yet; that is our next gate.

3 · What is actually unique?

Not offline forms and not phone colour-reading. It is the combination: capture that refuses false certainty, plus a server that will not close a case without a verified report, action, same-source retest and resident communication, plus residents in the same record. This is scoped to our dated desk review.

4 · What works today?

On synthetic data, on an emulator and on hosted staging: offline capture surviving crashes, sync without duplicates, the full guarded case loop (8/8 journey test), and the resident complaint and points APIs. Not yet: a physical phone, a real kit, the board UI in this repo, or resident app screens.

5 · How will users adopt it?

Through a shadow pilot. Workers keep their kit and lab. The app adds structure, not steps they do not already owe: a half-day of training, QR-tagged sources and a named follow-up owner. Hindi is still to come.

6 · What happens when it is wrong or unavailable?

Records save offline and queue. The manual reading is always available. The model is research-only, abstains, and has a remote kill switch. Lab results need a second person. During the pilot the existing process remains the record of decision.

7 · Who pays, and why?

Residential campuses first, from the estates or facilities budget, for proof of follow-up and less chasing time. Then NGO programmes. Residents never pay, and we never sell data. Willingness to pay is unvalidated. The first test is 10 buyer conversations and 2 letters of intent.

8 · How will impact be verified?

With pilot metrics against a 4-week shadow baseline: ownership within 24 h, days to referral, follow-through within 30 days, and evidence-less closures. We make no health-outcome claims. Those would need independent evaluation.

9 · What is the biggest weakness?

No real kit, lab partner, user or buyer yet: everything is synthetic, and ten production blockers are open. A free configurable tool may be "good enough" for some buyers.

10 · What will you prove next?

Physical-phone cold runs, a real kit protocol reviewed by a domain expert, a one-campus shadow pilot with baseline data, and a priced buyer conversation.

Field app and case workflow

Verified (synthetic)

code exists and passes automated tests or emulator runs on synthetic data

Partial

built with a named gap

Hackathon

planned for this event

Roadmap

future team scope

Proposed

suggestion, not a team commitment

FEATURE	USER	PROBLEM ADDRESSED	STATUS	EVIDENCE OR PLANNED VALIDATION
Field app				
Staff sign-in (Supabase Auth, ES256 verified)	Worker, supervisor	Only authorised people create records	Verified (synthetic)	Emulator sign-in against staging, app v2.2 (screenshot); provider tests
Source catalogue: search, QR allow-list, source history	Worker	Test is tied to the right source	Verified (synthetic)	43 Python + 22 Node tests; 10 hostile QR payloads rejected; emulator screens. Independent review owed
Kit protocol, lot/expiry eligibility, read-window timer (two-clock)	Worker	Results read outside the kit's window or with expired reagents	Partial	39 Node tests; runs on synthetic SYN-COLOR-001 only; no real kit yet
Guided capture with region marking	Worker	Unusable or unverifiable photos	Partial	Emulator only; reference-card locator missing; physical phone pending
Deterministic capture-quality checks and retake path	Worker	Guessing from bad photos	Verified (synthetic)	Fixtures 8/8, Kotlin 4/4; thresholds provisional
Manual reading with provenance, 'screening result only' wording	Worker	False certainty; screening mistaken for a lab result	Verified (synthetic)	12/12 review tests; wording tests; screenshots
On-device model suggestion with calibrated abstention	Worker	Reading variability	Partial	Research only: 68-parameter MLP, 896 B; 29/29 emulator parity, 1.7 ms median; synthetic data; not used in live flow
Crash-safe offline save with local receipt	Worker	Records lost when phones die or restart	Verified (synthetic)	Fault injection at commit boundaries; emulator kill -9 in airplane mode
Idempotent sync and ordered pull	Worker, supervisor	Duplicates and lost updates after reconnecting	Verified (synthetic)	Lost-response replay = duplicate, 1 row; PostgreSQL concurrent replay; emulator reconnect
Offline access lease, clock-rollback lock, revocation	Programme admin	Lost/stolen phones keep access offline	Verified (synthetic)	Emulator: 5 h clock rollback locked access, 0 records lost
Encryption at rest on the phone	All	Data exposure on a lost phone	Roadmap	Production blocker (T32); design choice pending
Hindi interface, TalkBack pass	Worker, resident	Language and accessibility barriers	Roadmap	T28 not started; static a11y fixes done
Case workflow				
Exactly one case per flagged sample	Supervisor	Flags that nobody owns	Verified (synthetic)	State-machine tests; journey step 3
Owner, lab referral, due date, overdue visibility	Supervisor	No deadline, no accountability	Verified (synthetic)	Case API tests (T17/T18), PostgreSQL race x5
Lab report recorded, then verified by a different person	Supervisor, lab reviewer	Unverified or self-approved results	Verified (synthetic)	Self-review refused at API and DB layer; journey step 5
Corrective action with completion evidence	Operator	'Fixed' with no proof	Verified (synthetic)	7/7 focused tests
Same-source, later retest linked to the case	Worker, supervisor	Closing without re-checking	Verified (synthetic)	g_retest guard tests
Resident communication record (never a delivery claim)	Supervisor	Residents never told	Verified (synthetic)	T22 tests; journey step 7-8
Evidence-checked closure with named missing items	Supervisor	Premature closure	Verified (synthetic)	Closure refused until verified report + action + retest + communication; closure policy awaits domain sign-off
Metrics and CSV export (formula-injection safe)	Programme lead	No view of overdue and repeat problems	Partial	14/14 tests; single-process job store
Supervisor web board	Supervisor	Needs a screen, not an API	Partial	API + CORS here; board is a separate app

Resident layer, platform, roadmap and proposals

FEATURE	USER	PROBLEM ADDRESSED	STATUS	EVIDENCE OR PLANNED VALIDATION
Resident layer				
Resident account (phone + password, lockout, forced change)	Resident	Anonymous, untraceable complaints	Partial	HTTP 12/12; phone ownership unverified (no SMS provider)
Complaint with reference number, rate limit; illness report alerts supervisors	Resident, supervisor	Complaints that go nowhere	Partial	Verified against the live database; no app screen yet; alert delivery not connected
Complaint linked to a case; +status lookup	Resident, supervisor	No feedback to the person who reported	Partial	review_complaint API; IVR intake needs telephony provider
Public source status (fuzzed location, never 'safe to drink')	Resident	False reassurance; privacy	Partial	API verified, map refresh by pg_cron observed; app portal shows demo fixtures
Points, rewards, sponsor redemptions, leaderboard	Resident, sponsor	Low reporting participation	Partial	Atomic, race-safe redemption proven; no app screen
Platform				
Tenant isolation across every /v1 route	All	One customer seeing another's data	Verified (synthetic)	Route matrix; cross-tenant ids look like missing ids. Blocker: API uses table-owner role, RLS off
Safe logging, request tracing, alert runbook	Operator	Leaked secrets; silent failures	Partial	Injected secrets never logged; alerts not wired to paging
Retention with deletion ledger that survives restore	Operator	Over-retention of photos and data	Verified (synthetic)	Interrupted purge resumes; restore reapplies deletions
Backup/restore, rollback with outbox replay, model kill switch	Operator	Unrecoverable failures	Partial	Rehearsed on staging; photo backup and scheduled dump missing
CI and hosted staging	Team	Regressions	Verified (synthetic)	CI green; staging smoke 7/7; /health/ready 200 on 25 Sep 2026
Demo				
5-minute live story: offline capture → refused closure → evidenced closure	Judges	Show the claim, not screens	Hackathon	Team demo plan; synthetic data visibly labelled
Roadmap				
Real kit protocol with domain reviewer; signed closure policy	Programme	Scientific and policy validity	Roadmap	T01/T46; needs a lab/domain partner
Validated kit-specific model on real preparations	Worker	Reading variability	Roadmap	T23-T27 on independent preparations, leave-phone-out tests
SMS / push / IVR delivery	Resident	Residents without the app	Roadmap	Providers B2-B4 not chosen
Shadow pilot and production release	Operator	Real-world proof	Roadmap	T38 protocol drafted; T39 checklist; 10 open blockers
Proposed				
Light shield; signed printable case receipt; kit-lot drift warning; voice instructions	Worker, resident	Capture quality, literacy, trust	Proposed	Team's own E01-E05, explicitly not committed
Lab turnaround tracker (referral → verified report, per lab)	Programme lead	Audits found 9-106 day delays	Proposed	Our suggestion; data already captured by case timestamps
Bridge the two case models (mobile-sync cases and v2 reports)	Team	Two parallel case records	Proposed	Our engineering recommendation before pilot

Source for every row: the team repository as of 25 Sep 2026 (current-state record, release evidence index, handoffs, tests)^[25]. Test counts are the team's recorded results; the author of this document did not re-run suites that write to the live database.

Every commercial and impact input, labelled **OBS** observed · **EXT** external benchmark · **TGT** team target · **ASM** illustrative assumption

INPUT	VALUE	LABEL	RATIONALE / EVIDENCE
USD→INR	₹96	ASM	FBIL reference 95.725 (11 Sep 2026), spot ~95.96 (25 Sep 2026), rounded
Campus Essentials price	₹7,500/month, billed yearly	ASM	Hypothesis; ~39% of ODK Cloud Standard (USD 199 ≈ ₹19,104/month); mWater platform is free, so price must be justified by enforced workflow + service
Campus Plus price	₹18,000/month	ASM	Hypothesis; larger campuses, lab-partner accounts, priority support
Programme price	₹30,000/month	ASM	Hypothesis; NGO / implementation-partner cluster up to ~300 sources
Setup fees	₹30,000 / ₹60,000 / ₹75,000	ASM	Source registration, QR tags, training day(s)
Onboarding delivery cost	₹15,000 / ₹30,000 / ₹60,000 per new customer	ASM	Staff days + travel + tags
Variable hosting/storage per customer	₹600 / ₹1,500 / ₹3,000 per month	ASM	Photos, backups, egress; SMS excluded (not built; would be pass-through)
Shared platform infrastructure	₹12,000/month Y1; ₹20k / 30k / 50k Y3	ASM	Repo cost register: USD 59/month pilot baseline, USD 70-100 provisional allowance; Supabase Pro USD 25; doubled for staging + backups
Year 1 opex (all scenarios)	₹7.2 L	ASM	Tools/legal/company ₹1.5 L, sales travel ₹1.5 L, kit + lab validation ₹2.0 L, independent security review ₹1.5 L, two free pilots ₹0.7 L
Founder pay, Year 1	₹0 (headline); ₹15 L sensitivity	ASM	Disclosed: unpaid founders flatter Year 1
Founder pay, Year 3	₹9 L each × 3	ASM	Sustainable staffing assumed from Year 3
Customer success staffing	₹4.8 L per 25 customers	ASM	Counted as direct delivery cost
Annual churn	20% / 15% / 10%	ASM	No renewal data exists
Flag rate (records needing follow-up)	5%	ASM	Orientation: 1.4-3.0% FTK-contaminated (Karnataka 2021-24, CAG) and 24% of household taps failing lab microbiology (FA 2024)
Baseline follow-through	18%	EXT	Karnataka 2023-24: 559 of 3,078 FTK-contaminated samples retested (CAG). Different context; our baseline must be measured in the pilot
Follow-through target	50% / 75% / 90%	TGT	Share of flagged results with verified lab result + retest within 30 days
Sources and test frequency	25 / 60 / 300 sources; monthly / monthly / quarterly	ASM	Per tier; to be replaced by pilot data
Residential share of colleges	30%	ASM	Not sourced; used only for the reachable-market illustration
Reachable geography share	10%	ASM	Direct sales in 2-3 states within three years

Definitions used

MRR = monthly recurring subscription revenue. ARR = MRR at a stated date × 12. Recognised revenue = subscription earned in the year (month-by-month in Year 1; linear build between start and end counts in Year 3) + setup fees. Gross profit = revenue – direct delivery costs (infrastructure, per-customer hosting, onboarding, customer success). Operating result = gross profit – operating expenses. Break-even = (opex + fixed infrastructure) ÷ annual contribution per customer at the Year-3 mix.

What each key number is, where it comes from, and its limits

CLAIM	REF	UNIT / DENOMINATOR	GEOGRAPHY	LIMITATION	USED
93.84 lakh FTK tests in 2024-25; 47.59 lakh in 2025-26 (to 12 Mar 2026); 24.80 lakh women trained (cumulative)	[1]	Water samples tested with FTKs; women trained, cumulative	India, rural (JJM)	State-reported to WQMIS; 2025-26 is a partial year, not a decline	Exec, p6
FTK results are indicative screening; adverse results need lab confirmation and corrective measures	[1]	Policy statement	India (JJM)	Describes the rule, not compliance with it	p3, p8
Karnataka: 0 of 4,540 (2021-22), 0 of 7,909 (2022-23) and 559 of 3,078 (2023-24) FTK-contaminated samples retested (18%)	[2]	Retested / contaminated FTK samples	Karnataka, state totals	State-level; one state; 7-district sample shows 549 of 910 (60%) retested in 2023-24 - different denominator	Exec, p4
Karnataka: lab turnaround 9-75 days (all 79 labs), 71-106 days (4 private empanelled) vs 24 h chemical / 48 h bacteriological norm	[2]	Days, sample receipt to report	Karnataka labs	Ranges, not medians	p4
Karnataka: remedial action not taken for 52% of contaminated samples (IMIS)	[2]	Share of contaminated samples	Karnataka	Period of IMIS chart not stated in text	p3
Karnataka: audit's own tests met the standard in only 2 of 28 test-checked villages	[2]	Villages	28 villages in 7 districts	Small purposive sample	p4
Kerala: district labs did not communicate results to District/Gram Panchayats for remedial action; WQMIS formats WQ1-6 lacked sampling date, report date, location and remedial action taken	[3]	Qualitative audit finding	Kerala, test-checked districts	Government disputes; says data uploaded to WQMIS	Exec, p3, p4
Kerala: average district-lab turnaround 45 / 40 / 33 days (Kozhikode / Kollam / Palakkad)	[3]	Days	3 test-checked districts	Government claims testing within norm; no records furnished	p4
76.0% of household tap samples passed microbiological tests; 72.8% at public institutions; 92.4% of households satisfied with quality; FTKs in 27.2% of villages; VWSC in 55.2%	[4]	Weighted % of samples / households / villages	19,401 Har Ghar Jal villages, 761 districts	Monsoon fieldwork (Jul-Oct 2024); only HGJ villages; third-party NABL lab	Exec, p6
2,870 water-quality testing labs (as on 4 Feb 2026); Citizen Corner shows village-level results	[5]	Labs	India	Count, not capacity or turnaround	p6, p7
Indore Bhagirathpura: 36 deaths reported 26 Dec 2025 - 25 Feb 2026; judicial commission reportedly linked 24 to sewage-contaminated municipal supply; six officials held responsible	[6]	Deaths	Bhagirathpura, Indore	Commission report confidential; findings reported via lawyers	p5
Indore: complaints to municipal helpline reportedly unattended for days from 25 Dec 2025	[8]	Qualitative	Bhagirathpura	Residents' accounts; not an official finding	p5
Kochi DLF: 441 residents ill in two weeks (health dept); lab report dated 29 May showed E. coli; residents allege it was not shared until 13 Jun	[10]	People ill	One apartment complex, Kakkanad	Disclosure delay is an allegation	p5
Chitradurga Kavadiarahatti: 4 of 5 water samples unfit; Vibrio cholerae found; tank served 220 houses; 5-6 deaths reported	[12]	Samples; deaths	One village	Death toll varied by report; cause of contamination disputed	p5
Gandhi Medical College hostel: 30+ of ~300 UG students ill; students say they had raised the tank issue earlier	[14]	Students ill	One campus	Student account; lab results not reported in source	p5
mWater Workflows support assignment, steps, automations and overdue alerts; platform free to end users; fees for services	[15]	Product capability	Global	Documentation read; no hands-on trial	p7
ODK Entities support longitudinal case management; offline entity updates need Collect/Central 2024.3+; ODK Cloud Standard USD 199/mo	[16]	Product capability, price	Global	Documentation read; no hands-on trial	p7, p16
45,473 colleges registered (AISHE 2021-22)	[19]	Colleges	India	Not all residential; residential share is an assumption	p15

The full ledger, with publisher, publication date and data period in separate columns, is delivered as *jalsakshi-source-and-assumptions-ledger.md*. IndiaSpend's Feb 2026 analysis repeats the FA-2024 figures, so it is not cited as independent corroboration.

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